Memory Is Not Enough: Separating Drift from Background Conflict in Continual Document Understanding

Anonymous CVPR submission

Paper ID *****

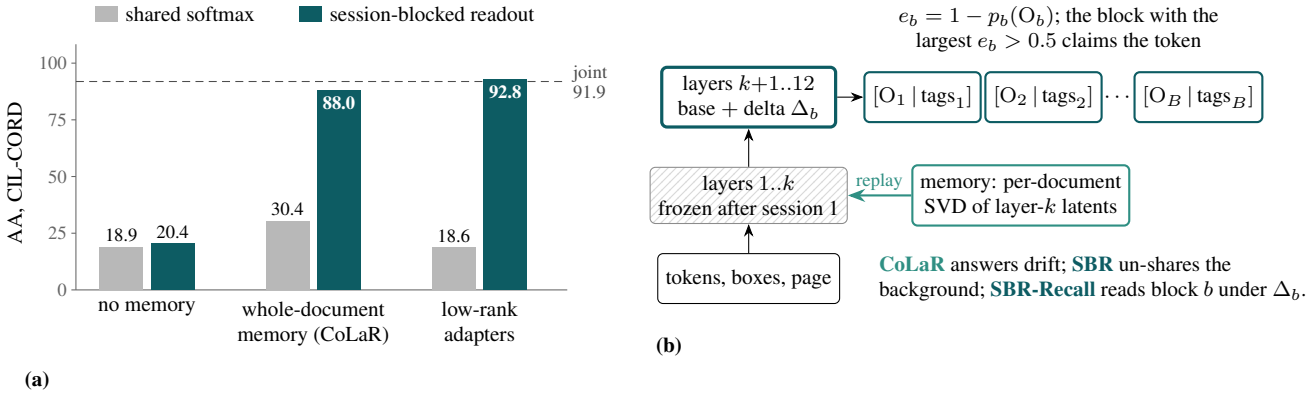


Figure 1. **Two causes of forgetting, and a remedy matched to each.** (a) Class-incremental CORD, LayoutLMv3, AA under session-restricted scoring; bars are the cell means of Tab. 2, which gives the seeds and protocol of each cell. A whole-document memory alone and a session-blocked readout alone both stay far below joint training; together they come within a few points of it, and one low-rank delta per session with a blocked readout (SBR-Recall) is on par with it without a memory, while a single low-rank adapter shared by all sessions with a shared softmax (our CL-LoRA port) collapses. (b) Layers 1..k are frozen after the first session; the memory stores per-document SVDs of layer-k activations and re-grounds the drifting upper layers (CoLaR); each session gets its own background logit and tokens are routed with no task identity (SBR); SBR-Recall evaluates each block under the trunk delta of the session that created it.

Abstract

Continual learning for document understanding is commonly treated as a single problem, catastrophic forgetting, addressed by a single remedy, replay. We show that continual key-information extraction instead fails for two separable reasons. The first is drift *of the output-facing computation, which arises in every scenario and on all four document encoders we study; freezing the affected layers relocates the drift rather than removing it, and none of the five method families we evaluate that update a shared network without old-task supervision recovers from it. The second,* background conflict*, arises only when the label space grows: current and stored documents supervise the shared background class inconsistently, an effect that no memory can repair and that the standard per-session evaluation conceals. A factorial analysis on class-incremental CORD separates the two causes: a whole-document memory and a session-blocked readout are each insufficient in isolation, whereas together they attain* 88.0 *average F1 against* 91.9 *for joint training. We propose a remedy for each cause.* ***CoLaR*** *(Compressed Latent Replay) replays a truncated SVD of each stored document's mid-network activations and matches raw-exemplar replay under domain shift without storing raw pixels or token identifiers.* ***SBR*** *(Session-Blocked Readout) assigns each session its own background logit and routes every token without task identity. Finally,* ***SBR-Recall*** *replaces the memory with per-session low-rank trunk updates when label growth itself supplies the routing key, and reaches* 92.8*, on par with joint training.*

1. Introduction

Organisations process a continuous stream of visually-rich documents (forms, invoices, receipts, contracts) whose meaning depends on the words and on where they sit on the page. Key-information extraction (KIE) is cast as token-level sequence labelling over text, 2-D layout and the page image, and multimodal encoders such as LayoutLMv3 [12], LiLT [37] and BROS [10] define the state of the art. Deployed systems are not static: new document types appear, schemas grow and the input drifts, while retraining on all history is blocked by cost and by privacy constraints on earlier data. This is continual learning (CL), whose central obstacle is catastrophic forgetting [24].

The CL toolbox (regularisation [14], distillation [18], replay [3, 32], prompts [34, 41], low-rank adapters [39]) treats forgetting as one phenomenon with one dial between stability and plasticity. On documents that picture produces a puzzle. Under a domain shift with a fixed label space, plain replay comes close to joint training on every backbone we test. When the label space grows, every method, replay included, stays far below joint training, and a better memory helps little. The usual reading is that class-incremental learning is simply a harder case of the same forgetting.

We show it is a different failure. Continual KIE fails for two *separable* reasons. **Drift**: the output-facing computation (upper layers and classifier) moves when it is trained on the next task, and on a fixed label space every method we tested that keeps updating a shared network without old-task supervision loses the old readout. **Background conflict**: under label growth a token labelled "O" means "none of the classes seen so far", so current documents label old entities as background while stored documents label new entities as background, and both push the same logit in opposite directions on the same surface forms. Memory addresses the first; no memory we tested resolved the second. The standard per-session scoring hides the second as well, because it charges a correct prediction of another session's entity as an error: a jointly trained model scores 33 under it and 91.9 once each error is charged once.

Figure 1a is the paper in one picture. On class-incremental CORD a whole-document memory (CoLaR, our compressed latent replay) with a shared softmax reaches 30.4 and a session-blocked readout (SBR) without memory reaches 20.4, both far closer to naive fine-tuning (18.9) than to joint training; the two together reach 88.0 ± 2.0, against 91.9 ± 1.3 for joint training. Neither remedy is sufficient alone and each cause needs its own remedy, which is what two separable causes predict.

Contributions.

1. **A corrected protocol for token-level class-incremental evaluation.** Session-restricted scoring charges every error at most once; we add a mask-free full-label F1, reported for clean-protocol runs as they complete, and show that the reported "floor" of class-incremental document IE is largely a scoring artefact (Sec. 3).
2. **A separation of two causes of forgetting.** Drift is present in every scenario, shared by four encoders, relocated rather than removed by freezing, and not repaired by five families of shared-network methods; background conflict is specific to label growth and survives a memory with perfect labels. A 2×2 over memory and readout isolates them (Sec. 4).
3. **SBR and SBR-Recall.** The Session-Blocked Readout (SBR) gives each session its own background logit and routes every token with no task identity; with a whole-document memory it comes within a few points of joint training on class-incremental CORD and reaches it on a mixed stream. SBR-Recall stores a low-rank trunk delta per session and reads each block under its own delta, which removes drift by construction and needs no memory on a single-schema stream (Secs. 5.2 and 5.3).
4. **CoLaR** (Compressed Latent Replay)**.** A falsification ladder shows that a document memory must keep whole documents, and that each document can be compressed on its own: per-document truncated SVD of mid-network activations matches raw-exemplar replay under domain shift at 60 MB (Sec. 5.1).

2. Related Work

Continual-learning families. Following Wang et al. [38], remedies fall into regularisation (EWC [14], SI [44], MAS [1], LwF [18]), replay (GEM [21], ER [32], DER++ [3], iCaRL [31]), optimisation geometry (GPM [33], OGD [5], C-Flat [2]), methods over a frozen backbone (L2P [41], DualPrompt [40], CODA-Prompt [34], O-LoRA [39], CL-LoRA [9], RanPAC [25]) and architectural isolation [22, 43]. Replay is empirically the strongest [23, 38] but stores raw data. Exemplar-free alternatives replay class-conditional feature statistics (PASS, FeTrIL, FeCAM [6, 28, 52]) or merge models and realign the classifier (LCA [35]).

Class-incremental learning as prediction plus task identification. Kim et al. [13] decompose class-incremental learning into within-task prediction and task-identity prediction, and tie the latter to out-of-distribution detection; recent adapter methods add one module per task and need no task identity at test time, either by combining the features of all modules (EASE [51]) or by folding them into the pre-trained weights (InfLoRA [19]). SBR is an instance of the decomposition that predicts task identity explicitly, per token, through the per-session background logit. It has one property specific to token-level extraction: every document of a session contains tokens that are true negatives for that session's classes,

so this logit is a task-identity detector trained with real negatives, at no extra cost. In continual NER, ICE [20] trains per-session classifiers on a frozen encoder and SpanKL [48] uses per-type span heads with distillation; neither replays stored examples.

Latent and compressed replay. REMIND [8] stores product-quantised mid-level CNN feature maps for each training image and replays them through the plastic upper layers. Latent Replay [27] stores each example's activations at a chosen layer, and generative replay [36] samples hidden representations from one class-conditional generator. Like CoLaR, REMIND and Latent Replay keep one entry per example, but each of their examples carries a single label. A document carries one label per token, bound to a position, so the stored entry must keep that binding; CoLaR compresses each document with a truncated SVD instead of product quantisation (Sec. 5.1).

Where forgetting lives; continual IE. Ramasesh et al. [30] localise forgetting in image classifiers to the layers nearest the output, and Zhai et al. [45] find that multimodal LLMs forget under fine-tuning even when the image encoder is frozen; our drift analysis agrees with the former and adds the intervention. Continual NER handles the shift of "O" with distillation and pseudo-labels (CPFD [46]), causal-effect distillation with curriculum learning (CFNER [50]) or entity-triplet generation [42]; continual relation extraction uses episodic memory with prototypes [4, 7, 49]. All are sentence-level and text-only. Document-level continual learning is nascent: GDI-Bench [17] limits forgetting in multimodal-LLM fine-tuning by freezing most parameters, chosen by their update magnitude; continual table detection [26] uses plain ER; and ProtoNER [16], the closest work to ours, adds new key-information classes to a LayoutLM-family model from a few labelled documents with prototypical networks. None of these document-level works separates the causes of forgetting.

3. Setup

Task and scenarios. A document is a triple of OCR tokens, normalised boxes and the page image; the model emits BIO tags per token and is scored by entity-level F1 (`seqeval`). We use FUNSD (149/50 train/test forms), CORD (800/100/100 receipts, 30 classes) and SROIE (626/347 receipts). **DIL**: FUNSD→SROIE→CORD under one fixed four-entity schema; only the input distribution shifts. **CIL-CORD**: CORD's 30 classes arrive six at a time over five sessions; the head grows, and tokens of classes not yet introduced are labelled "O". **Mixed**: six sessions alternating new classes within CORD with new domains (FUNSD, SROIE) and ending in a FUNSD revisit. Details are in the supplement (Sec. A).

Backbones and methods. LayoutLMv3-base [12] is the primary encoder; LiLT [37], BROS [10] and text-only BERT are secondary. Heads are linear. The comparison set is fixed in advance: naive fine-tuning and joint training as references, and one published method per family: ER [32] and DER++ [3], LwF [18], CODA-Prompt [34], CL-LoRA [9], LCA [35], and the continual-NER methods CPFD [46] and IS3 [29]. Replay baselines keep 200 raw documents (DER++ $\alpha{=}\beta{=}0.5$); in clean-protocol cells ER instead keeps as many raw documents as our memory. LwF uses $T{=}2$. Our CL-LoRA port trains a single rank-8 adapter on the attention projections of the frozen encoder, shared by all sessions, unlike the published one-adapter-per-task design, so its numbers describe this port. Our memory keeps 50 documents per session (per task on DIL).

Training protocol. AdamW, learning rate $5{\times}10^{-5}$, weight decay 0.01, gradient clipping 1.0, batch size 2, three seeds. Under the *clean protocol* every session trains in full precision for at most 10 epochs (joint training: 100) with early stopping, patience 2, on a 10% split held out from that session's *training* data; the test split is never used for model selection. Cells marked † come from earlier runs (the *legacy protocol*), which differ from the clean protocol in three ways. Every legacy cell early-stops on the session's *test* split, except joint training on CIL-CORD and Mixed, which holds out 10% of its pooled training set. On CIL-CORD and Mixed, epoch caps differ by method (the DIL caps are in the supplement): 100 for naive, joint training, ER, DER++, LwF, CPFD and IS3, 30 for CODA-Prompt and CL-LoRA, 10 for the CoLaR-family rows and LCA. Precision differs too: most re-scored CIL-CORD and Mixed cells ran in fp16 mixed precision, but IS3 and LCA always ran in fp32, and seed 42 of ER, DER++, LwF, CPFD, CL-LoRA and CODA-Prompt on CIL-CORD ran in fp32, so several cells mix precisions across seeds; the CIL-CORD cells of CoLaR-SBR, no-memory SBR, SBR-Recall and LexSlot, and the DIL SBR cells, are fp32. In Tab. 2 the shared-softmax cells are therefore fp16 or mixed and the blocked-readout cells fp32. The DIL rows of our first grid (Tab. 1) saved no configuration; their settings are the grid defaults and could not be confirmed per run. The supplement (Sec. D) lists the settings of every † cell. A cell never combines the two protocols.

Scoring. In CIL-CORD and the class-incremental halves of Mixed, session j's test labels show only session j's classes; every other entity is "O". Scoring raw predictions against them (the practice we call legacy scoring) charges a model that *correctly* labels another session's entity with a false positive in every other session, so a jointly trained

model reads 33. *Session-restricted scoring* maps a predicted tag outside session j's label set to "O" before span matching, without re-taking the argmax among the allowed tags. Each predicted entity is then scored in the one session that owns its class, so an error is charged at most once and a confusion between sessions still costs a miss. Because a session's test split omits receipts with none of its entities, we also report a mask-free *full-label F1*: the final model on the unmasked CORD test set with all 30 classes. Only clean-protocol runs record it, so Tab. 3 gives it for the runs completed so far, with the number of seeds marked. From $R[i, j]$ (F1 on session j after training session i) we report average accuracy $\text{AA} = \frac{1}{T}\sum_j R[T, j]$ and backward transfer $\text{BWT} = \frac{1}{T-1}\sum_{j<T}(R[T, j] - R[j, j])$.

4. Two Causes of Forgetting

4.1. Cause 1: drift of the output-facing computation

Forgetting is total within one boundary, on every backbone. Under naive sequential fine-tuning on the stream FUNSD→CORD→SROIE the first task collapses after one further task: FUNSD falls from 89.1 to 3.3 on LayoutLMv3, 83.9 to 2.1 on LiLT and 88.9 to 6.6 on BROS. Table 1 gives the DIL picture on four backbones.

Where it sits. Representations drift cumulatively with depth: per-token CKA [15] between consecutive checkpoints is 1.00 at the embeddings, 0.78 over the early layers, 0.34 in the middle, 0.25 at the last layer and 0.16 at the classifier, with the same profile on all four backbones (Fig. 2, left) and under LayoutLMv3 modality masks the full, text+layout and text-only models forget comparably (BWT -89.8 and -87.2 over three seeds, -79.1 over the two live text-only seeds; the model without text forgets least, -66.7 over three seeds). The classifier is also where parameters that mattered for the old task are overwritten: old-task-Fisher-weighted displacement is two to three orders of magnitude larger there than in any encoder bucket of the document encoders. That weighting favours the head by construction, and inside the encoder the weighted displacement sits in the early and middle layers, not the late ones, so we read "late layers" as a statement about representations only (supplement, Sec. D). This agrees with the image-classification anatomy of Ramasesh et al. [30] and is not our contribution; the interventions below are.

Freezing relocates the drift. Freezing the whole network after the first task gives AA 41.1 with BWT 0.0 on DIL: nothing is forgotten and nothing is learned. Freezing only the head and late layers, with the early and middle encoder left plastic, gives AA 38.8 and BWT -76.8, the naive level, while early-layer CKA, $\approx$1.0 under naive training, falls to 0.19. The drift moved into whatever remained plastic (Fig. 2, right; seed 42).

Table 1. **Drift and its remedy are shared by four encoders** (DIL, AA, mean $\pm$ s.d. over three seeds, all cells legacy protocol†, from our first grid, whose per-run settings were not saved). On average EWC and LwF stay within ten points of naive training and at least thirty points below joint training (the reference) on every backbone, whereas replay comes within one point of joint training or exceeds it. "–": did not fit the 6 GB GPU.

Backbone	Naive	EWC	LwF	ER	DER++	Joint
LayoutLMv3	41.3±1.5	41.2±1.1	43.7±1.1	87.9±1.4	88.0±0.9	88.7±1.1
LiLT	40.1±0.5	48.4±5.9	–	85.7±0.5	84.4±0.3	83.8±1.2
BROS	40.3±1.8	45.0±4.9	35.1±3.1	87.8±0.4	87.5±1.1	87.5±1.3
BERT	39.4±0.3	46.6±3.0	40.5±0.1	77.1±0.9	77.4±0.6	77.2±0.4

Only old-task supervision of the readout repairs it. On average replay comes within one point of joint training, or exceeds it, on all four backbones; EWC and LwF stay within 12 points of naive training and at least 27 below joint training in every run (Tab. 1). We then built or ported five families that keep updating a shared network with no stored old-task targets (head merging, merging with Gaussian classifier realignment [35], subspace transfer, isolated head slots, feature-Gaussian replay): head merging is a no-op (21.6 against 21.9 unmerged at its operating point), merging with realignment reaches 43.9, subspace transfer 39.8, head slots 42.2, and feature-Gaussian replay 66.0 by recovering the densely labelled task only, against 41.3 for naive and 88 for replay (supplement, Sec. C). Where we measured the mechanism, on naive training and LwF, the head's output marginal on old tasks snaps to the just-trained task's marginal (cosine 0.997–0.998 against 0.60 to the old task's own), and classes that are never re-exercised fall to exactly 0.0 F1. Five post-hoc corrections of that marginal, each with a pass bar fixed before the run, recover at most -0.1 AA, because old-class probability mass is $\sim 10^{-6}$. Our claim is scoped accordingly: on a fixed label space, no method we tried that updates a shared trunk and readout without old-task supervision recovers the old tasks. Section 5.3 shows the claim does not extend to label growth, where isolation becomes possible.

The size of a remedy's benefit is backbone-dependent; the cause is not. The gap between joint and naive training on Mixed agrees to 2.5 points across the four backbones, whereas the benefit of EWC over naive spans 17 points and changes sign (-11.9 on LayoutLMv3, -2.6 on LiLT, $+3.5$ on BROS, $+5.2$ on BERT; legacy scoring, supplement Sec. D). A single-backbone comparison of methods without old-task supervision can land on either side of zero, which is why Tab. 1 reports four.

4.2. Cause 2: background conflict under label growth

On CIL-CORD under legacy scoring naive training, LwF, ER and DER++ sit at 16–19 AA on average against 33 for joint



Figure 2. **Drift is output-facing, and freezing moves it.** Left: linear CKA between checkpoints before and after the second task under naive fine-tuning falls with depth on all four backbones (1.0 = no drift; mean over three seeds). Right: freezing the head and late layers leaves backward transfer unchanged, (a) whatever the number of frozen layers and (b) on three DIL task orders and a ten-session CIL-CORD stream (LayoutLMv3, seed 42, no memory).

Table 2. **Memory × readout on CIL-CORD** (LayoutLMv3, AA, session-restricted scoring; mean ± s.d. over three seeds, per-seed values when fewer; †legacy protocol, settings per cell in the supplement). Rows change how drift is handled, columns how the background is supervised. The third row uses low-rank adapters and no memory, but its two cells differ in adapter design as well as in readout: our port of CL-LoRA [9] (one rank-8 adapter on the attention projections of the frozen pre-trained encoder, shared by all sessions, trained with distillation and an orthogonality penalty against its own earlier states; fp16 or fp32 by seed, 30-epoch cap) with a shared softmax, and SBR-Recall (one rank-64 delta per session on the attention and feed-forward layers above k, base frozen after the first session, fp32, 10-epoch cap) with the blocked readout.

	Shared softmax	**Session-blocked readout**
No memory	$18.9\pm0.3^{\dagger}$	$20.4^{\dagger}$
Whole-document memory (CoLaR)	$28.8/31.9^{\dagger}$	$\mathbf{88.0\pm2.0}^{\dagger}$
Low-rank adapters, no memory	$18.7/18.6^{\dagger}$	$\mathbf{92.8\pm0.5}^{\dagger}$
Joint training (reference)	$91.9\pm1.3^{\dagger}$	

training, and several baselines collapse below that band; this has been read as a floor. Session-restricted scoring removes the artefact for the reference, which rises to 91.9 ± 1.3, and leaves the methods far below it: ER 24.3 ± 2.4, CoLaR 30.4, naive 18.9 ± 0.3 (Tab. 3). The gap that remains is real, and it is not a memory problem.

A document stored at session t carries "O" on every entity introduced after t, and a current document carries "O" on every entity of earlier sessions. With one shared softmax both sets of labels supervise the same background logit, on receipts that look alike, in opposite directions. Improving the memory does not remove the contradiction: refreshing the stored labels with the true current labels, a ceiling no deployable method can reach, gives 61.2 (seed 42), and restricting the replay loss to the classes alive at storage time gives 42.0 (two seeds). Perfect memory labels recover about half of the gap, because current documents still mislabel old entities. The continual-NER methods designed for the shift of "O" do not recover it either (Tab. 3).

4.3. The two causes separate

Table 2 crosses the remedy for each cause. Without a memory, blocking the readout barely helps (18.9 to 20.4): the labels are consistent but the trunk drifts. With a whole-document memory and a shared softmax, drift is contained but the labels conflict (30.4). Only the cell with both approaches joint training (88.0 ± 2.0 against 91.9 ± 1.3). The third row uses low-rank adapters and no memory. CL-LoRA, one low-rank adapter shared by all sessions, with a shared softmax, collapses (18.6, although its sessions reach 81 to 95 F1 when they are learned), and SBR-Recall, one low-rank delta per session, with a blocked readout, reaches 92.8 ± 0.5. This row is not a controlled swap of the readout: its two cells differ in adapter design (one shared adapter against one delta per session, with different rank, placement and training; Tab. 2) as well as in readout. Drift can be handled by memory or by per-session deltas; in every cell we ran, the background conflict was resolved only at the readout.

5. Remedies

5.1. CoLaR: a memory for drift

A memory repairs drift by re-exercising the readout on inputs that keep their old labelling. To find the least it must store we vary the stored object and hold the mechanism fixed: activations are captured at the output of layer k, layers $1..k$ are frozen after the first task, and stored activations are replayed through the plastic upper layers and head (DIL, LayoutLMv3; single-seed diagnostics unless stated).

Whole documents retain; summaries collapse. Replaying the stored activations of 50 whole documents per task retains AA 87.3. Summaries of the *same* store collapse to the naive level: a rank-16 pooled spectral summary 41.9, per-class Gaussians 39.4, real k-means centroids 36.7–39.7. A single-variable control isolates the ingredient: four whole documents retain 63.8, and the identical token features placed on layout carriers from other documents retain 36.7. The binding of a token's feature to its position and its label is worth 27 points. The requirement is specific to documents.

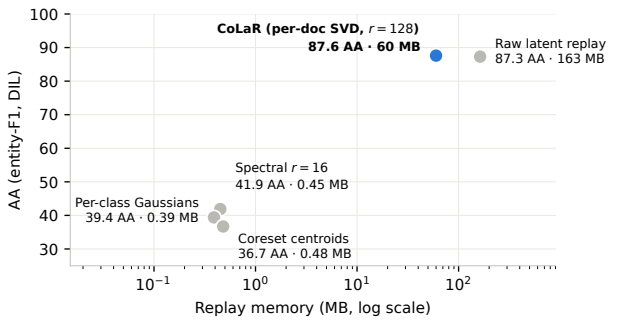


Figure 3. Memory size versus retention (DIL, LayoutLMv3, AA, seed 42; log-scale memory). Per-document SVD (CoLaR) matches raw latent replay at 2.7× less storage; every summary that breaks whole-document binding falls to the naive level.

On ViT-B/16 image class-incremental learning (CIFAR-100, ImageNet-R; supplement, Secs. B and C) a class-conditional Gaussian head (SLCA [47], 87.5/65.0) beats the same latent memory (80.0/55.1), and a ridge readout refit on the trained trunk reaches 88.3/72.3: there the trunk is intact and only readout statistics are stale. An image carries one label, so a per-class statistic summarises its readout; a document carries one label per token bound to a position, and the Gaussian analogue is null on documents in every form we tried (39.4 as a store).

Per-document compression. The pooled feature space is close to full rank, but one document's tokens share a layout and a vocabulary, so its activation matrix $H_j \in \mathbb{R}^{N_j \times d}$ (N_j tokens, width d) is low-rank. CoLaR (Compressed Latent Replay) stores a rank-r truncated SVD per document, $H_j \approx U_r \Sigma_r V_r^\top$, with the document's boxes, attention mask and labels, rebuilds $\hat{H}_j$ at replay, and injects it at layer k; the loss is cross-entropy on the current batch plus cross-entropy on replayed documents. At k=4, r=128 and 50 documents per task it retains 87.8±0.5 AA on DIL in 60 MB (1 MB = 10^6 bytes throughout), against 87.9±1.4 for ER with 200 raw documents and 88.7±1.1 for joint training, and is 2.7× smaller than the uncompressed latent memory it matches (87.3 at about 163 MB; Fig. 3); r=64 gives 80.6 at about 32 MB. The memory holds no raw pixels or token ids. We make no formal privacy claim: mid-network activations can leak content. Rebalancing the memory (class-balanced loss, coverage-based selection, entity-weighted compression) trades one task for another without moving AA, and read-side additions are negative (supplement, Sec. C).

5.2. SBR: a readout for the background conflict

Let session b introduce the tag set $\mathcal{T}_b$. A shared softmax normalises over $\{\mathrm{O}\} \cup \bigcup_b \mathcal{T}_b$. The Session-Blocked Readout (SBR) gives every session its own background logit $o_b(x) = w_b^\top h(x)$ on the token feature $h(x)$ and normalises each block separately:

$$p_b(\cdot \mid x) = \mathrm{softmax}\big([\, o_b(x),\, z_{\mathcal{T}_b}(x)\,]\big), \tag{1}$$

where $z(x)$ are the tag logits of the usual linear head and $z_{\mathcal{T}_b}(x)$ its entries for the tags in $\mathcal{T}_b$. Within block b a token labelled "O" in session-b data is truly background, so the block's supervision is never stale. For a token x with label y in a document of session s,

$$\mathcal{L}(x, y, s) = \sum_b m_{sb}\, \mathrm{CE}\big(p_b(\cdot \mid x),\, y_b\big), \qquad y_b = \begin{cases} y & y \in \mathcal{T}_b \\ \mathrm{O}_b & \text{otherwise,} \end{cases} \tag{2}$$

with one rule for the mask: m_{sb}=1 when session s annotates every tag of $\mathcal{T}_b$ (the block is supervised fully); m_{sb}=1 with y_b=O_b when b belongs to a different annotation schema, so none of its entities can occur in the document; and m_{sb}=0 otherwise, because the document's "O" labels say nothing about block b. Stored documents follow the rule of the session that stored them. A session that adds no tags creates no block, so on a fixed label space SBR has a single block.

Inference needs no task identity. Each block reports how strongly it claims the token, $e_b = 1 - p_b(\mathrm{O}_b \mid x)$; the block with the largest e_b labels the token with its best tag if $e_b > 0.5$, and the token is "O" otherwise. The trunk is shared across blocks and kept stable by the CoLaR memory.

5.3. SBR-Recall: isolation in place of memory

With SBR the memory is written during training and never read at test time. SBR-Recall also stores, for each session $b > 1$, a low-rank delta Δ_b [11] of the layers above k (rank 64 on the attention projections and feed-forward layers), freezes these layers after the first session (the base θ_0; Δ_1=0), and computes block b under the trunk state that learned it:

$$p_b(\cdot \mid x) = \mathrm{softmax}\big([\, o_b,\, z_{\mathcal{T}_b}\,]\big)\big(x;\, \theta_0 + \Delta_b\big). \tag{3}$$

Every term of Eq. (2) is computed under the delta its block is read under, at training and at test time. Old blocks can then not drift, since neither the base nor their delta is updated, and routing by e_b is unchanged. Inference costs one pass through the upper layers per session.

SBR-Recall is the blocked-readout cell of the adapter row of Tab. 2. A shared-softmax counterpart would have to combine or merge the deltas at test time (as EASE and InfLoRA do with their modules) or predict the session with a separate model; we did not run one. SBR's per-session background logit is such a predictor, trained on real negatives at no extra cost. It works without a memory because label growth supplies what isolation normally lacks, a routing key: each

Table 3. **Comparison on LayoutLMv3** (AA under session-restricted scoring on CIL-CORD and Mixed; mean ± s.d. over three seeds, per-seed values when fewer; "–": not run). Full-label F1 is the final model's F1 on the unmasked CORD test set, recorded by clean-protocol runs only; $^{n=k}$: k clean seeds so far. ER keeps 200 raw documents in † cells and as many as CoLaR in clean cells. LexSlot: extra head slots selected by OCR-vocabulary overlap (see Ablations). †legacy protocol (test-split early stopping except CIL-CORD and Mixed joint training; epoch cap and precision differ by row, see Sec. 3). ^{c}LwF only: one fp16 seed collapsed to all-background (CIL-CORD seed 123, Mixed seed 7) and is excluded; LwF did not collapse in its fp32 run. CPFD and IS3 keep their zero seeds, which are not precision effects (CPFD also collapses in fp32; IS3 always ran in fp32). ^{b}with the 50-document memory; 44.7 without.

	CIL-CORD		Mixed	DIL
Method	AA	full-label F1	AA	AA
Naive	18.9±0.3†	–	44.4/43.7†	41.3±1.5†
ER	24.3±2.4†	–	75.6/73.3†	87.9±1.4†
LwF	18.4/18.5†,c	–	47.7†,c	43.7±1.1†
DER++	20.4±1.4†	–	75.1/74.0†	88.0±0.9†
CODA-Prompt	14.0/15.9†	–	12.1/10.6†	32.5±1.2†
CL-LoRA (2025)	18.7/18.6†	–	45.7/43.9†	41.1±0.3†
CPFD (2023)	2.7/0.0†	–	0.3/15.1†	6.1±7.3†
IS3 (2024)	5.6/7.7†	–	0.0/40.6†	38.3†
LCA (2026)	13.7/7.0†	–	38.0/36.4†	43.9±2.5†
CoLaR	28.8/31.9†	–	75.6/74.6†	87.8±0.5†
CoLaR-SBR	88.0±2.0†	88.8/91.2$^{n=2}$	**89.8/89.6**†	86.2±1.3†
+ LexSlot head slots	89.7±0.5†	–	89.6±1.0†	87.0±0.4†
SBR-Recall	**92.8±0.5**†	–	85.9†,b	–
Joint training (reference)	91.9±1.3†	–	86.9/88.5†	88.7±1.1†

session's background logit, trained on that session's true negatives, tells the model which delta a token belongs to. Where there is no such key, isolation does not replace memory. On a fixed label space there is one block and nothing to route by. Across schemas the all-background terms of Eq. (2) have no positive counterweight without stored documents: on Mixed, SBR-Recall without a memory scores 44.7 (seed 42), with other-schema blocks silent until their domain returns, and 85.9 with the 50-document memory restored.

6. Experiments

Main comparison. Table 3 reports the fixed comparison set. On CIL-CORD every continual method with a shared softmax stays at or below CoLaR (30.4), and the two blocked readouts close most or all of the gap: CoLaR-SBR 88.0±2.0 and SBR-Recall 92.8±0.5 against 91.9±1.3. On Mixed, CoLaR-SBR reaches 89.7 against 87.7 for joint training and 74.5 for ER. On DIL, where only drift is present and SBR has one block, CoLaR, CoLaR-SBR and ER all come close to joint training on average (87.8, 86.2, 87.9 against 88.7); SBR (86.2) is below CoLaR (87.8) there, and its only differences are the separate background logit and the thresholded decision.

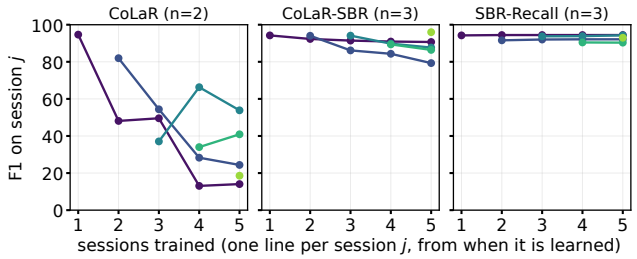


Figure 4. **Retention per session on CIL-CORD** (LayoutLMv3, session-restricted scoring; F1 on session j after each later session, mean over the n seeds given in each panel title). With a shared softmax every session decays as soon as the next arrives; the blocked readout holds them, with a slow loss on the earliest sessions; under SBR-Recall every curve is flat.

Ablations. The blocked readout on a trunk frozen after the first session scores 14.8 (later sessions cannot be learned), and without a memory 20.4, so neither the readout nor plasticity alone is the effect. Tuning the claim threshold and a per-block bias on the test set itself changes AA by at most 0.6 (seed 42), so what SBR still loses to joint training is retention of early sessions (Fig. 4), which is the part SBR-Recall removes. For SBR-Recall without a memory, rank 8 on the attention projections alone gives 82.5 (seed 42) with flat retention but two sessions stuck at 65 and 68 F1: isolation is sound and the capacity of the delta is what CORD's look-alike price classes need. Stacking LexSlot on CoLaR-SBR (extra head slots that a session updates only when its documents' OCR vocabulary selects them) leaves the mean within two points on all three scenarios and narrows the spread between seeds on CIL-CORD and DIL; both mechanisms partition the same readout. Unlike CoLaR, LexSlot stores the token ids of its documents.

Cost. CoLaR-SBR stores 50 compressed documents per session (101 MB on CIL-CORD) and one background logit per session. SBR-Recall stores no documents on a single-schema stream and 57 MB of half-precision deltas for the four later sessions, and pays one upper-layer pass per session at inference (about 3.7× the compute of one forward pass at five sessions and k=4).

7. Limitations

The study covers three English datasets with test sets of 50 to 347 documents and four encoder-only backbones; SBR and SBR-Recall are evaluated on LayoutLMv3. Cells marked † early-stop on the test split (except CIL-CORD and Mixed joint training). Every cell is marked by its protocol and never mixes the two, so a comparison across protocols is visible from the marks in the tables; the supplement lists the settings of every † cell. The freezing experiment and

the memory-content ladder are single-seed diagnostics; the tables are three-seed unless a cell shows fewer values. Joint training is a reference, not a bound: with early stopping it can be matched or slightly exceeded. SBR-Recall grows by one delta and one inference pass per session, freezes its base on the first session, so its accuracy can depend on which session comes first, and does not replace memory on a fixed label space or across annotation schemas; its Mixed result is one seed. SBR's supervision rule uses the annotation schema of each session, which is known at training time and is not used at test time. CoLaR is a replay method whose memory grows with the number of sessions; its settings were chosen on LayoutLMv3 and transfer to BROS but trail raw replay by 5–9 points on average on the text-heavy BERT and LiLT. CoLaR stores no raw pixels or token ids, which is weaker than privacy.

References

[1] Rahaf Aljundi, Francesca Babiloni, Mohamed Elhoseiny, Marcus Rohrbach, and Tinne Tuytelaars. Memory aware synapses: Learning what (not) to forget. In *European Conference on Computer Vision (ECCV)*, 2018. 2

[2] Ang Bian, Wei Li, Hangjie Yuan, Chengrong Yu, Mang Wang, Zixiang Zhao, Aojun Lu, Pengliang Ji, and Tao Feng. Make continual learning stronger via C-Flat. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 7608–7630, 2024. 2

[3] Pietro Buzzega, Matteo Boschini, Angelo Porrello, Davide Abati, and Simone Calderara. Dark experience for general continual learning: A strong, simple baseline. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. 2, 3

[4] Li Cui, Deqing Yang, Jiaxin Yu, Chengwei Hu, Jiayang Cheng, Jingjie Yi, and Yanghua Xiao. Refining sample embeddings with relation prototypes to enhance continual relation extraction. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL-IJCNLP)*, 2021. 3

[5] Mehrdad Farajtabar, Navid Azizan, Alex Mott, and Ang Li. Orthogonal gradient descent for continual learning. In *Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2020. 2

[6] Dipam Goswami, Yuyang Liu, Bartłomiej Twardowski, and Joost van de Weijer. FeCAM: Exploiting the heterogeneity of class distributions in exemplar-free continual learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 2

[7] Xu Han, Yi Dai, Tianyu Gao, Yankai Lin, Zhiyuan Liu, Peng Li, Maosong Sun, and Jie Zhou. Continual relation learning via episodic memory activation and reconsolidation. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020. 3

[8] Tyler L. Hayes, Kushal Kafle, Robik Shrestha, Manoj Acharya, and Christopher Kanan. REMIND your neural network to prevent catastrophic forgetting. In *European Conference on Computer Vision (ECCV)*, 2020. 3

[9] Jiangpeng He, Zhihao Duan, and Fengqing Zhu. CL-LoRA: Continual low-rank adaptation for rehearsal-free class-incremental learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 30534–30544, 2025. 2, 3, 5

[10] Teakgyu Hong, Donghyun Kim, Mingi Ji, Wonseok Hwang, Daehyun Nam, and Sungrae Park. BROS: A pre-trained language model focusing on text and layout for better key information extraction from documents. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 10767–10775, 2022. 2, 3

[11] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022. 6

[12] Yupan Huang, Tengchao Lv, Lei Cui, Yutong Lu, and Furu Wei. LayoutLMv3: Pre-training for document AI with unified text and image masking. In *Proceedings of the 30th ACM International Conference on Multimedia (MM)*, pages 4083–4091, 2022. 2, 3

[13] Gyuhak Kim, Changnan Xiao, Tatsuya Konishi, Zixuan Ke, and Bing Liu. A theoretical study on solving continual learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. 2

[14] James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A. Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, et al. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences (PNAS)*, 114(13):3521–3526, 2017. 2

[15] Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, 2019. 4

[16] Ritesh Kumar, Saurabh Goyal, Ashish Verma, and Vatche Isahagian. ProtoNER: Few shot incremental learning for named entity recognition using prototypical networks. In *Business Process Management Workshops (BPM 2023 International Workshops)*, pages 70–82. Springer, 2024. 3

[17] Siqi Li, Yufan Shen, Xiangnan Chen, Jiayi Chen, Hengwei Ju, Haodong Duan, Song Mao, Hongbin Zhou, Bo Zhang, Bin Fu, Pinlong Cai, Licheng Wen, Botian Shi, Yong Liu, Xinyu Cai, and Yu Qiao. GDI-Bench: A benchmark for general document intelligence with vision and reasoning decoupling. *arXiv preprint arXiv:2505.00063*, 2025. 3

[18] Zhizhong Li and Derek Hoiem. Learning without forgetting. In *European Conference on Computer Vision (ECCV)*, 2016. 2, 3

[19] Yan-Shuo Liang and Wu-Jun Li. InfLoRA: Interference-free low-rank adaptation for continual learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 23638–23647, 2024. 2

[20] Minqian Liu and Lifu Huang. Teamwork is not always good: An empirical study of classifier drift in class-incremental information extraction. In *Findings of the Association for Computational Linguistics: ACL*, 2023. 3

[21] David Lopez-Paz and Marc'Aurelio Ranzato. Gradient episodic memory for continual learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. 2

[22] Arun Mallya and Svetlana Lazebnik. PackNet: Adding multiple tasks to a single network by iterative pruning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018. 2

[23] Marc Masana, Xialei Liu, Bartłomiej Twardowski, Mikel Menta, Andrew D. Bagdanov, and Joost van de Weijer. Class-incremental learning: Survey and performance evaluation on image classification. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(5):5513–5533, 2023. 2

[24] Michael McCloskey and Neal J. Cohen. Catastrophic interference in connectionist networks: The sequential learning problem. *Psychology of Learning and Motivation*, 24:109–165, 1989. 2

[25] Mark D. McDonnell, Dong Gong, Amin Parvaneh, Ehsan Abbasnejad, and Anton van den Hengel. RanPAC: Random projections and pre-trained models for continual learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 2

[26] Mohammad Minouei, Khurram Azeem Hashmi, Mohammad Reza Soheili, Muhammad Zeshan Afzal, and Didier Stricker. Continual learning for table detection in document images. *Applied Sciences*, 12(18):8969, 2022. 3

[27] Lorenzo Pellegrini, Gabriele Graffieti, Vincenzo Lomonaco, and Davide Maltoni. Latent replay for real-time continual learning. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2020. 3

[28] Grégoire Petit, Adrian Popescu, Hugo Schindler, David Picard, and Bertrand Delezoide. FeTrIL: Feature translation for exemplar-free class-incremental learning. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2023. 2

[29] Shengjie Qiu, Junhao Zheng, Zhen Liu, Yicheng Luo, and Qianli Ma. Incremental sequence labeling: A tale of two shifts. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 777–791, 2024. arXiv:2402.10447. 3

[30] Vinay V. Ramasesh, Ethan Dyer, and Maithra Raghu. Anatomy of catastrophic forgetting: Hidden representations and task semantics. In *International Conference on Learning Representations (ICLR)*, 2021. 3, 4

[31] Sylvestre-Alvise Rebuffi, Alexander Kolesnikov, Georg Sperl, and Christoph H. Lampert. iCaRL: Incremental classifier and representation learning. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017. 2

[32] David Rolnick, Arun Ahuja, Jonathan Schwarz, Timothy P. Lillicrap, and Greg Wayne. Experience replay for continual learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019. 2, 3

[33] Gobinda Saha, Isha Garg, and Kaushik Roy. Gradient projection memory for continual learning. In *International Conference on Learning Representations (ICLR)*, 2021. 2

[34] James Seale Smith, Leonid Karlinsky, Vyshnavi Gutta, Paola Cascante-Bonilla, Donghyun Kim, Assaf Arbelle, Rameswar Panda, Rogerio Feris, and Zsolt Kira. CODA-Prompt: Continual decomposed attention-based prompting for rehearsal-free continual learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. 2, 3

[35] Tung Tran, Danilo Vasconcellos Vargas, and Khoat Than. LCA: Local classifier alignment for continual learning. In *International Conference on Learning Representations (ICLR)*, 2026. 2, 3, 4

[36] Gido M. van de Ven, Hava T. Siegelmann, and Andreas S. Tolias. Brain-inspired replay for continual learning with artificial neural networks. *Nature Communications*, 11:4069, 2020. 3

[37] Jiapeng Wang, Lianwen Jin, and Kai Ding. LiLT: A simple yet effective language-independent layout transformer for structured document understanding. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022. 2, 3

[38] Liyuan Wang, Xingxing Zhang, Hang Su, and Jun Zhu. A comprehensive survey of continual learning: Theory, method and application. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 46(8):5362–5383, 2024. 2

[39] Xiao Wang, Tianze Chen, Qiming Ge, Han Xia, Rong Bao, Rui Zheng, Qi Zhang, Tao Gui, and Xuanjing Huang. Orthogonal subspace learning for language model continual learning. In *Findings of the Association for Computational Linguistics: EMNLP*, 2023. 2

[40] Zifeng Wang, Zizhao Zhang, Sayna Ebrahimi, Ruoxi Sun, Han Zhang, Chen-Yu Lee, Xiaoqi Ren, Guolong Su, Vincent Perot, Jennifer Dy, and Tomas Pfister. DualPrompt: Complementary prompting for rehearsal-free continual learning. In *European Conference on Computer Vision (ECCV)*, 2022. 2

[41] Zifeng Wang, Zizhao Zhang, Chen-Yu Lee, Han Zhang, Ruoxi Sun, Xiaoqi Ren, Guolong Su, Vincent Perot, Jennifer Dy, and Tomas Pfister. Learning to prompt for continual learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022. 2

[42] Yawen Yang, Fukun Ma, Shiao Meng, Aiwei Liu, and Lijie Wen. GenCNER: A generative framework for continual named entity recognition. In *2025 International Joint Conference on Neural Networks (IJCNN)*, pages 1–9, 2025. arXiv:2510.11444. 3

[43] Jaehong Yoon, Eunho Yang, Jeongtae Lee, and Sung Ju Hwang. Lifelong learning with dynamically expandable networks. In *International Conference on Learning Representations (ICLR)*, 2018. 2

[44] Friedemann Zenke, Ben Poole, and Surya Ganguli. Continual learning through synaptic intelligence. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017. 2

[45] Yuexiang Zhai, Shengbang Tong, Xiao Li, Mu Cai, Qing Qu, Yong Jae Lee, and Yi Ma. Investigating the catastrophic forgetting in multimodal large language model fine-tuning. In *Proceedings of the Conference on Parsimony and Learning (CPAL)*, pages 202–227, 2024. 3

[46] Duzhen Zhang, Wei Cong, Jiahua Dong, Yahan Yu, Xiuyi Chen, Yonggang Zhang, and Zhen Fang. Continual named

entity recognition without catastrophic forgetting. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023. 3

[47] Gengwei Zhang, Liyuan Wang, Guoliang Kang, Ling Chen, and Yunchao Wei. SLCA: Slow learner with classifier alignment for continual learning on a pre-trained model. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 19091–19101, 2023. 6

[48] Yunan Zhang and Qingcai Chen. A neural span-based continual named entity recognition model. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 13993–14001, 2023. 3

[49] Kang Zhao, Hua Xu, Jiangong Yang, and Kai Gao. Consistent representation learning for continual relation extraction. In *Findings of the Association for Computational Linguistics: ACL 2022*, 2022. 3

[50] Junhao Zheng, Zhanxian Liang, Haibin Chen, and Qianli Ma. Distilling causal effect from miscellaneous other-class for continual named entity recognition. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 3602–3615, 2022. 3

[51] Da-Wei Zhou, Hai-Long Sun, Han-Jia Ye, and De-Chuan Zhan. Expandable subspace ensemble for pre-trained model-based class-incremental learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 23554–23564, 2024. 2

[52] Fei Zhu, Xu-Yao Zhang, Chuang Wang, Fei Yin, and Cheng-Lin Liu. Prototype augmentation and self-supervision for incremental learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021. 2